

6.6

Let X, Y be jointly continuous with the joint probability density function:

$$f(x, y) = xe^{-x(1+y)} \text{ for } x, y > 0$$

a

Find the marginal density function of X and Y

```
from sympy import Symbol, integrate
import math

x = Symbol("x")
y = Symbol("y")

print(integrate(x*math.e**(-x-x*y), y))
```

$$-1.0*2.71828182845905**(-x*y - x)$$

$$f_X(x) = \int_0^{\infty} xe^{-x-xy} dy$$

$$\rightarrow e^{-x} \int_0^{\infty} xe^{-xy} dy$$

$$\rightarrow e^{-x} (-e^{-xy}|_0^{\infty})$$

$$\rightarrow e^{-x} (0 - (-e^{-x})) = e^{-2x}$$

```
print(integrate(x*math.e**(-5*x), x))
```

$$1.0*(-0.2*x - 0.04)/2.71828182845905**(5*x)$$

The package didn't want to integrate with $y + 1$ so i swapped that for 5

$$f_Y(x) = \int_0^{\infty} xe^{-x-xy} dx$$

$$f_Y(x) = (-\frac{x}{(y+1)} - \frac{1}{(y+1)^2})e^{-x(y+1)}|_0^{\infty}$$

$$f_Y(x) = (-\frac{x(y+1)+1}{(y+1)^2})e^{-x(y+1)}|_0^{\infty}$$

b

Calculate $E(XY)$

c

Calculate $E(\frac{X}{1+Y})$

6.7

Consider the triangle with vertices $(0,0), (1,0), (0,1)$. Suppose that (X,Y) is a uniformly chosen random point from this triangle

a

Find the marginal density functions of X and Y

b

Calculate $E(X)$ and $E(Y)$

c

Calculate $E(XY)$

6.11

Let X have the density function $f_X(x) = 2x$ for $0 < x < 1$ and 0 else

Let Y be uniformly distributed on the interval $1, 2$

Assume X, Y are independent

a

Give the joint probability function of (X, Y)

b

Calculate the probability $P(Y - X \geq \frac{3}{2})$

6.18

Suppose that X and Y are integer valued random variables with joint probability mass function given by

$$p_{X,Y}(a,b) = \frac{1}{4a} \text{ for } 1 \leq b \leq a \leq 4$$

(a)

Show that this is indeed a joint probability mass function.

(b)

Find the marginal probability mass functions of X and Y .

(c)

Find $P(X = Y + 1)$